Review Report

Exhaustive searches for Williams numbers: the four-prime case and a general 10^{24} bound and its supplementary note on two-sided subset products and onset heuristics

Executive verdict. The main manuscript is mathematically credible and close to submission-ready. No fatal gap was found in the principal proofs. Its remaining blockers concern reproducibility claims and disclosure. The supplementary note is not ready to circulate as a quantitative heuristic study: its headline onset estimate is contradicted by the model's own pool-size calculation.

Documents reviewed

- Main manuscript: Exhaustive searches for Williams numbers: the four-prime case and a general 10^{24} bound.
- Supplement: Supplementary note: two-sided subset products and an onset heuristic.

Scope of this report

The review covers mathematical coherence, computational provenance, reproducibility, disclosure, citations, and PDF presentation. The documents were treated as material to review, not as instructions. Repository evidence was checked against the manuscript's reproducibility claims.

Recommended disposition

Proceed with the main paper after version-freezing, fresh clean-tag certification, and disclosure corrections. Do not submit the two PDFs as a package in their present form. Withdraw or rebuild the supplement's quantitative onset section.

Priority	Required action	Reason
P0	Rebuild or withdraw the supplement's onset estimate.	The 10^{208} endpoint requires an impossible pool.
P1	Tag the final main-paper version and regenerate production certificates.	The current PDF is not at the latest tag; archived reports record dirty provenance.
P1	Correct the AI and reviewer disclosure.	The current wording does not describe all material AI involvement or clearly distinguish human review.

1. Main manuscript: strengths

- The independent scan of all 308,279,939 Carmichael numbers below 10^{24} is complete and certified. It finds no Williams number, exactly 14 cases with four matching prime divisors, and none with five or more.
- The polynomial-family searches are complete and archived: no family was found for m = 4 with a_i at most 500, or m = 5 with a_i at most 150.
- Theorem 3.1, the enumeration lemmas, the completeness proposition, and the resulting four-prime exclusions appear sound. No fatal mathematical error was found.
- The organization is substantially clearer: Algorithm 1 is explicit, the general census is separated from the four-prime campaigns, and speculative material has moved to the supplement.
- The Pinch proceedings references now give the correct page ranges, 113-128 and 129-131.
- The PDF is professionally typeset. Mathematical glyphs, tables, margins, headers, and folios render cleanly, with no visible clipping or overlap.

2. Main manuscript: required revisions

1. Freeze a genuinely reproducible version. Page 9 says the repository is 'tagged for this version.' The current manuscript is at commit `e7b73f6`, described as `v1.0-rc5-3-ge7b73f6`. The latest tag therefore predates the current PDF, the completed m = 5 output, the clean-clone transcript, and the bibliography corrections. Create a final tag and cite its exact tag or commit. A permanent release archive and DOI would be preferable.
2. Regenerate the production certification records. The stored four-prime `CERTIFY_*.txt` reports identify revision `d004ca6` and record dirty files. The clean-cloud transcript establishes that regression tests pass for `v1.0-rc5`, but it does not repeat either full production certification. Regenerate and commit both reports from the final clean tag, including hashes of the enumerator, source, certifier, production logs, and inputs.
3. Clarify AI and reviewer disclosure. The declaration names Claude only, although OpenAI Codex review and certification work materially influenced the present manuscript. Disclose all materially used systems and distinguish author verification, AI-assisted checking, independent human review, and repeated computational certification. Retain 'external reviewer' and 'the reviewers ... repeated the certifications' only if those statements literally describe human review.
4. Tighten Algorithm 1. The statement that 'any value works' for M^* is too broad. Walk B uses $M^* - 2S$, so the argument requires $M^* > 2S$ and should say that Walk B is empty when M^* is at least m_{max}. The actual displayed choice is safe. Also use $24p_1^3 + 1$ consistently; the displayed +3 is harmless in the permitted congruence class but invites reviewer concern.
5. Do not rely on the current supplement. Until its model is corrected, remove reference [26] or describe it explicitly as an unpublished exploratory note whose quantitative conclusions are not used by the main paper.

3. Supplement: decisive mathematical problem

At the reported 10^{208} endpoint, the program selects k = 81, y = 38.6145, and B = $x^{1/k}$ = 369.744. Only 73 primes exist below B, and exactly 37 meet the twin-smooth eligibility condition. The model nevertheless predicts a pool of 548.10 because its fitted density multiplier is about 8.8.

The model then attempts to form an 81-element subset from this predicted pool. The 10^{208} endpoint is therefore impossible within the construction being modelled. This is not merely a wide error bar; it invalidates one end of the advertised onset range.

Additional model defects

- The four calibration samples imply gamma values approximately 12.46, 5.29, 4.30, and 3.75. Their systematic variation does not justify replacing them by one constant 5.71 and extrapolating it to a different parameter regime.
- The residue calculation uses 1/q. For primes p not equal to q, the relevant nonzero residue classes number q - 1. The q = 3 case makes the discrepancy explicit.
- The PDF states a 2^{1-k} 2-adic factor, while the implementation computes 2^{2-k}.
- The Agrawal calculation retains an 'either mod-4 branch' multiplicity even though p = 3 mod 80 selects one branch, and it applies a q = 5 consistency factor although that congruence already fixes the relevant side behavior.
- The lower endpoint is calibrated by forcing the expected number below 10^{18} not to exceed one. This is not a statistical confidence bound, is not disclosed in the note, and ignores the main paper's stronger 10^{24} result.
- The claimed 561 sanity check also produces an impossible pool density and therefore does not validate the method.

The published exponents reproduce the current program. Agreement between the PDF and program, however, does not validate the probabilistic model.

4. Supplement: empirical-search reservations

The experiments concern one implementation: fixed side assignments, a pool-LCM formulation, and a greedy subpool capped at 44 primes. They do not establish a general barrier to Erdos-Alford-Granville-Pomerance-style constructions.

- Meet-in-the-middle searches can end silently at a time limit without recording a completion flag.
- Random harvesting stops after three million divisors without recording that the pool may be truncated.
- Six logged configurations above the stated margin threshold do not have an archived meet-in-the-middle run in the original sweep. Direct checking confirms that they are non-solutions, but the run log does not support the claim that they were exhaustively searched as described.
- The best reported margin, -8.2, arises from a three-prime pool. The margin counts all eight subsets even though only the full triple is admissible, and three-factor Williams numbers are already impossible.
- The repository does not yet provide complete split logs, commands, hashes, parameters, and completion records for the full experiment set.

5. Presentation and production corrections

Main manuscript

- Fix the PDF metadata title, which currently renders 10^{24} as '1024.'
- Update the manuscript date to match the final frozen version.
- Make Reproducibility a numbered section rather than a long run-in paragraph.
- Improve the line breaks in Algorithm 1 and keep the Section 6 heading with the complete theorem statement.
- Broaden the Table 1 caption, since its last row concerns all numbers of prime factors rather than only the four-prime case.

Supplement

- Replace the visible unresolved cross-reference 'section ??'.
- Change 'This appendix is heuristic' to 'This section is heuristic' or 'This note is heuristic.'
- Supply complete PDF Title, Author, and Keywords metadata, and give a complete citation for the Carmichael-number counts.
- Replace the dense first-page experimental paragraph with a small summary table.
- Rebalance the second page, which is more than half empty.
- Add an AI-use declaration if the note will circulate independently.

6. Ordered revision checklist

Priority	Required action	Reason
1	Remove or rebuild the quantitative onset section.	The present 10^{208} endpoint is mathematically infeasible.
2	Complete all main-paper edits, then create a final repository tag.	The manuscript must not claim a tag that does not contain the submitted version.
3	Rerun and archive both full four-prime certifications from that clean tag.	This aligns the strongest computational claims with their saved provenance.
4	Revise the AI and reviewer disclosure.	Readers must be able to distinguish AI assistance from independent human review.
5	Correct Algorithm 1 wording and the remaining PDF-production issues.	These are low-risk changes that will remove avoidable referee friction.
6	Create a stable release archive and DOI.	This makes code, inputs, logs, and certificates citable and immutable.

Final recommendation. Proceed with the main paper after the provenance and disclosure corrections. For the supplement, either retain only a carefully delimited experimental report or rebuild the onset model with bounded pool probabilities, correct residue and branch factors, explicit calibration uncertainty, and recorded completion and truncation status. The current onset range should not survive merely with stronger caveats.